



PROJECT DOCUMENTATION

EVENTFORCE MANAGEMENT SYSTEM

USING SALESFORCE

Salesforce | Record-Triggered Flow | Approval Process | Apex

TEAM DETAILS

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1. Introduction

Project Overview

Event planning companies handle many things at the same time: client requests, event scheduling, venue reservations, vendor assignments, cancellations and feedback. When these live in notebooks, chats and spreadsheets, bookings clash, vendors are missed and clients are not kept informed. The EventForce Management System brings all of this into one Salesforce CRM, where every client, event, venue, vendor and feedback record sits in a single connected data model.

The application is built in a Salesforce Developer Edition org. Custom objects hold the data, a validation rule keeps contact data clean, a Record-Triggered Flow reminds clients before their event, an Approval Process controls cancellations, Apex stops double bookings and keeps venue status current, and reports with a dashboard give the team a live view of upcoming events. Access is controlled with profiles, roles, a permission set, organization-wide defaults and a sharing rule.

Purpose

- Centralize client, event, venue, vendor and feedback information in one connected CRM.
- Streamline event booking and venue reservation, and prevent a venue from being booked twice on the same date.
- Update venue availability automatically when an event is confirmed or cancelled.
- Remind clients three days before their event without manual follow-up.
- Route cancellation requests through a manager approval with email alerts.
- Give each type of user only the access that their job needs.
- Show upcoming events through reports and a dashboard for better decisions.

2. Ideation Phase

Problem Statements

The table below records the problems that motivated the project, written from the point of view of each person affected.

#	I am	I am trying to	But	Which makes me feel
PS-1	An event operations manager	Manage event planning efficiently	Client, event, venue and vendor information is hard to manage centrally	Frustrated
PS-2	An event coordinator	Create and manage bookings	There is no central platform and many steps are manual	Overwhelmed
PS-3	A venue manager	Manage venue availability	The same venue may be assigned to several events	Uncertain
PS-4	A vendor manager	Coordinate vendors for events	Vendor details and assignments are managed separately	Overwhelmed
PS-5	An event coordinator	Send reminders on time	Reminders are handled manually	Stressed
PS-6	An event administrator	Manage cancellation requests	They need a controlled approval and status updates	Frustrated

#	I am	I am trying to	But	Which makes me feel
PS-7	An event coordinator	Manage client feedback	It must stay linked to both the event and the client	Concerned
PS-8	An event operations manager	Monitor operations and performance	Data is not organized into reports and dashboards	Less confident

Summary of needs. Scattered information, double-booked venues, missed vendors, manual reminders, uncontrolled cancellations and missing reports all point to one need: a single connected system that checks bookings, updates venue status, reminds clients, routes cancellations for approval and limits access by role.

Empathy Map

Persona: Event Coordinator. The map captures what the coordinator experiences while planning events.

THINKS & FEELS Wants every booking to be correct and every client to feel looked after.	HEARS Client requests, change requests and vendor calls arriving all day.	SEES Event dates, venues, event status, client details and upcoming events.
SAYS & DOES Schedules events, talks to clients and raises cancellation requests.	PAINS Double bookings, manual reminders, missed vendors and unclear cancellation approval.	GAINS Clash-free bookings, automatic venue status, automatic reminders and a clear view of upcoming events.

Brainstorming and Idea Prioritization

Four approaches were considered, and ideas were then placed on an impact-versus-feasibility grid.

Idea	Approach	Key limitation
Manual registers	Write event, client, venue and vendor details by hand	Time-consuming and hard to maintain
Spreadsheet tracking	Track bookings in spreadsheets	Limited automation and no real-time coordination
Standalone portal	Build a dedicated event portal	High development and maintenance effort
Salesforce CRM (chosen)	Centralize event operations on Salesforce	Needs proper configuration, which this project provides

	LOW FEASIBILITY	HIGH FEASIBILITY
HIGH IMPACT	High impact, low feasibility - Client portal for self-service - SMS / WhatsApp reminders	High impact, high feasibility - Block double booking of venues - Auto-update venue availability 3-day client reminder - Manager approval for cancellations - Reports and dashboard

LOW IMPACT	Low impact, low feasibility • Mobile app for coordinators	Low impact, high feasibility • Custom tabs for each object • Event Planner Lightning App • Email format validation • Lookup filter on Feedback
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Top prioritized ideas: double-booking prevention, automatic venue status, client reminder Flow, cancellation approval, and reports with a dashboard on upcoming events.

Final Selection

Salesforce CRM was chosen because it offers:

- Automation: reminders, approvals and status updates through Flow, Approval Process and Apex
- Scalability: it supports more events, clients, vendors, venues and users
- Real-time visibility through reports and dashboards
- Built-in security with profiles, roles, permission sets and sharing settings
- Less manual follow-up for the whole team

3. Requirement Analysis

Customer Journey Map

An event moves through eight stages, from the moment a client asks for a service to the feedback collected afterwards.

Stage	What happens in Salesforce	Outcome
1. Enquiry and client registration	Client requirements and contact details are captured.	Client and event recorded
2. Planning and booking	Event type, date, budget and status are created and managed.	Event planned
3. Venue selection	Availability is checked and the venue is reserved without double booking.	No clash
4. Vendor coordination	Vendors and services are assigned to the event through Event Vendor.	Vendors assigned
5. Confirmation and communication	Confirmed events trigger the reminder email to the client.	Client informed
6. Execution and monitoring	Progress is followed through records, reports and the dashboard.	Clear visibility
7. Cancellation and approval	Requests go through the approval process with status updates and alerts.	Controlled cancellation
8. Feedback and closure	Feedback is captured and past events are marked Completed.	Event closed, insights gathered

If an event must be cancelled, the status is set to Pending Cancellation, the record is locked and a manager approves or rejects the request. An approved cancellation sets the status to Canceled and frees the venue again.

Solution Requirements

Functional requirements

FR No	Requirement
FR-1	Event management: create and manage Event records
FR-2	Client management
FR-3	Venue management
FR-4	Vendor management
FR-5	Event and vendor assignment (many-to-many through Event Vendor)
FR-6	Venue availability and double-booking prevention
FR-7	Automated event reminders (3 days before the event)
FR-8	Event cancellation approval with email alerts
FR-9	Feedback management linked to both the event and the client
FR-10	Reports and dashboards
FR-11	Event status automation (venue status and past events marked Completed)
FR-12	Data migration through the Data Import Wizard

Non-functional requirements

Attribute	Description
Usability	The Event Planner Lightning App puts all tabs, reports and dashboards in one navigation bar
Security	Profile-, role- and permission-based access control
Reliability	Accurate automation, approvals and venue booking validation; rules run on save
Performance	Efficient processing of event data; the batch job processes past events in batches of 200 records
Availability	Cloud-based Salesforce access for authorized users
Scalability	Supports more events, clients, vendors, venues and users
Maintainability	Flows, Apex, validation rules and security can be monitored and updated from Setup
Data integrity	Validation rules and controlled data migration keep records accurate

Data Flow

From the client request to the final report, data moves through the following stages:



Supporting flows. Venue booking validation: event saved, Apex trigger checks venue and date, then an error is shown or the venue status is updated. Event completion: event date passes, the scheduled batch runs, status is set to Completed. Cancellation path: Event Status = Pending Cancellation, record locked, Manager Review, Approved or Rejected, email alert to the event owner.

Technology Stack

Layer	Technology
Platform	Salesforce CRM, Developer Edition org
Data	Custom objects Event, Client, Vendor, Venue, Feedback and Event Vendor (junction); Lookup and Master-Detail relationships; formula field; lookup filter
Logic	Apex class, Apex triggers, Batch Apex and Schedulable Apex (API version 67)
Automation	Record-Triggered Flow (scheduled path), Approval Process, Email Alerts, Field Updates, validation rule
UI	Custom tabs, Event Planner Lightning App, Reports and Dashboards
Security	Profiles, Roles, Users, Permission Set, Organization-Wide Defaults, Sharing Rule
Data loading	Salesforce Data Import Wizard

4. Project Design

Problem-Solution Fit

Problem	Solution	Benefit
Information is scattered	Custom objects for Event, Client, Venue, Vendor, Feedback and Event Vendor	Central, structured data
Venue booked twice on the same date	PreventDoubleBooking trigger and VenueStatusHelper	Venue schedule stays accurate
Vendor tracking is hard	Event Vendor junction object	Many-to-many assignment
Invalid client emails	Email_Valid_Address validation rule	Cleaner contact data
Late client reminders	Record-Triggered Flow with a scheduled path (3 days before)	Reminders without manual work
Uncontrolled cancellations	Event_Cancellation_Process approval with record lock	Manager approval and status updates
Past events left open	BatchCompleteEvents run by ScheduleCompleteEvents	Events closed automatically
Feedback not linked	Feedback with Event and Client lookups and a lookup filter	Organized feedback
No visibility	Reports and EventForce Operations Dashboard	Quick view of upcoming events
Different access needs	Profiles, roles, permission set, OWD and sharing rule	Secure, controlled access

Proposed Solution

Parameter	Description
Problem	Event details live in notebooks, chats and spreadsheets, so bookings clash, vendors are missed and clients are not kept informed.
Solution	A Salesforce CRM that stores clients, events, venues, vendors and feedback in one connected data model.
Innovation	Apex rules stop double bookings and update venue status automatically, while a scheduled Flow and an approval process remove manual follow-up.
Impact	Fewer booking errors, less manual follow-up, cleaner data and a clear view of upcoming events.
Business model	Event planning companies working with clients, venues, vendors, coordinators and admins.
Scalability	Scales through Salesforce automation, with room for time-slot checks, more approval steps, vendor performance tracking and integrations such as SMS or WhatsApp.

Solution Architecture

The solution is organized in layers. Users work in the Event Planner app, automation and Apex apply the business rules, and Salesforce records hold the data.

Users	Client Event Coordinator Vendor Manager Venue Manager Event Admin
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Presentation layer	Event Planner Lightning App Custom tabs (Events, Venues, Clients, Vendors, Feedbacks) New Events with Venue Report Upcoming Events by Month EventForce Operations Dashboard
Automation layer	Client Reminder - 3 Days Before (Flow) Event_Cancellation_Process (Approval) Email Alerts Field Updates Email_Valid_Address (Validation Rule)
Apex layer	EventTrigger13 VenueStatusHelper PreventDoubleBooking BatchCompleteEvents ScheduleCompleteEvents
Data layer	Event__c Client__c Vendor__c Venue__c Feedback__c Event Vendor (junction)
Security layer	Profiles Roles Permission Set Organization-Wide Defaults Sharing Rule

5. Project Planning and Scheduling

Planning Approach

- Agile methodology with sprint-based execution (four sprints)
- Work is split into epics and user stories and sized in story points
- Velocity is calculated from completed story points and used to check that the team stayed on track

Product Backlog and Sprint Schedule

Sprint	Epic	Story	User story / task	Points	Priority	Owner
1	Event Management	US-01	As an Event Coordinator, I want to create events with type, date and status so that event information is standardized.	3	High	Naveen P
1	Client and Venue Management	US-02	As an Event Coordinator, I want to associate a client and venue with an event so that coordination is centralized.	5	High	Saravanan K
2	Vendor Management	US-03	As a Vendor Manager, I want to assign multiple vendors to an event so that services can be coordinated.	5	High	Roopan G
2	Notifications and Automation	US-04	As a Client, I want timely event reminders so that I know when my event is approaching.	3	High	Mukil Doss S
3	Approval Management	US-05	As an Event Coordinator, I want cancellation requests to follow an approval process so that cancellations are controlled.	5	High	Naveen P
3	Venue Management and Apex	US-06	As a Venue Manager, I want double bookings prevented so that venue schedules remain accurate.	8	High	Saravanan K
4	Security and Access Control	US-07	As an Event Administrator, I want role-based access so that users see only appropriate information.	5	High	Roopan G
4	Reports and Dashboards	US-08	As Management, I want reports and dashboards so that event operations can be monitored.	5	Medium	Mukil Doss S

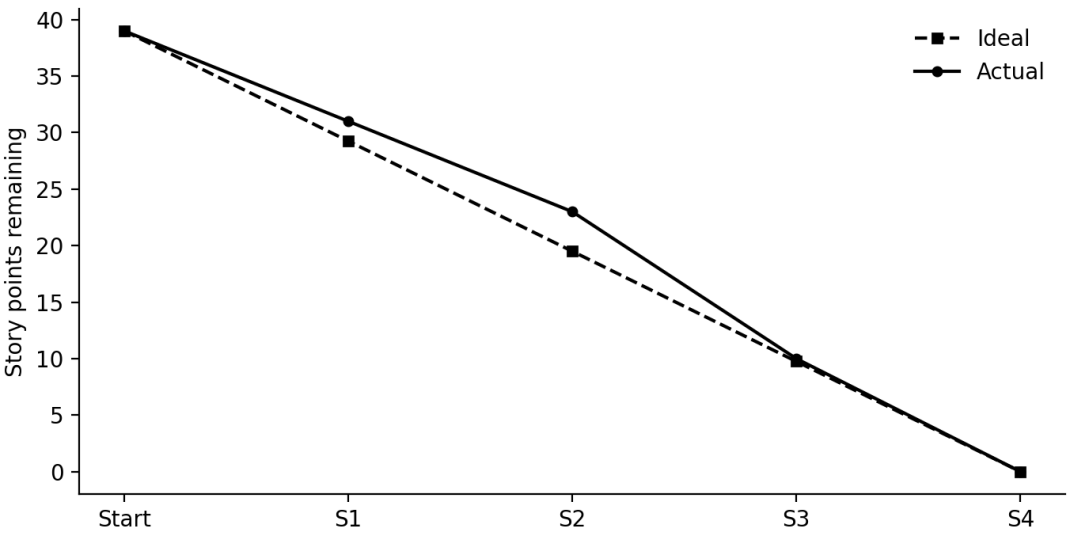
Story-point total: 39 across four sprints (8 + 8 + 13 + 10).

Sprint Tracker and Velocity

Sprint	Story points	Start date	End date	Points completed	Release date
Sprint-1	8	02 Oct 2026	03 Oct 2026	8	03 Oct 2026
Sprint-2	8	02 Oct 2026	03 Oct 2026	8	03 Oct 2026
Sprint-3	13	02 Oct 2026	03 Oct 2026	13	03 Oct 2026
Sprint-4	10	02 Oct 2026	03 Oct 2026	10	03 Oct 2026

Average velocity = total story points / number of sprints = 39 / 4 = about 9.8 points per sprint. Sprint 3 carried the most work (13 points) because the Apex double-booking logic and the approval process were the largest parts of the project.

Sprint Burndown (story points remaining)



Sprint burndown: ideal (dashed) versus actual (solid).

6. Project Development

The topics covered while building the solution were org setup, data modeling, validation, automation with Flow and Approval Process, Apex, the Lightning App, reports and dashboards, data loading and the Salesforce security model.

Milestone 1: Salesforce Developer Org

A Salesforce Developer Edition org was signed up for using the developer sign-up page (first name, last name, email, role Developer, company, country India, postal code and a unique username in the form name@organization.com). The account was verified from the email link, a password and security question were set, and the org opened in Setup. All objects, automation and code in this project were built inside this org.

Milestone 2: Custom Objects

Five main objects were created from Setup > Object Manager > Create > Custom Object: **Event, Client, Vendor, Venue and Feedback**. A sixth object, **Event Vendor**, is a junction object. For each object, Allow Reports, Allow Search and the Notes and Attachments related list were enabled and the object was Deployed. Event also has Allow Activities and Track Field History. Feedback uses an Auto Number record name with the display format F-{0000} starting at 0.

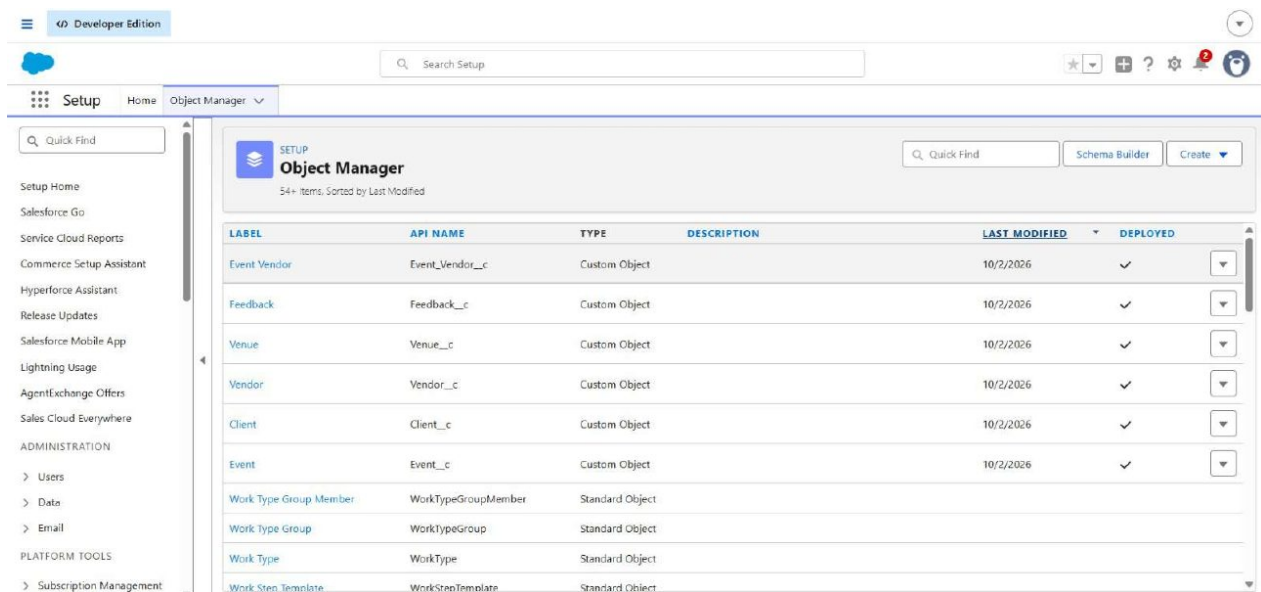


Figure 1. Object Manager showing the custom objects Event Vendor, Feedback, Venue, Vendor, Client and Event.

Milestone 3: Fields and Relationships

Each object holds the fields it needs. The tables and screenshots below show the final field set.

Object	Key fields
Event	Event Name (Text 80), Client (Lookup), Venue (Lookup), Event Date (Date), Event Type (Picklist), Event Status (Picklist), Event Budget (Formula, Currency)
Client	Client Name, Phone, Email, Address (Text Area 255), City (Picklist), Country (Picklist)
Vendor	Vendor Name, Phone, Email, Service Type (Multi-select Picklist), Status (Picklist)

Object	Key fields
Venue	Venue Name, Address (Text Area 255), Location (URL), Capacity (Number), Availability Status (Picklist)
Feedback	Feedback (Auto Number), Client (Lookup), Event (Lookup), Rating (Picklist), Comments (Long Text Area)
Event Vendor	Event (Master-Detail), Vendor (Master-Detail)

Picklist values

Field	Values
Event Type	Wedding, Corporate, Birthday, Anniversary, Festival, Concert, Other
Event Status	Planned, Confirmed, Completed, Pending Cancellation, Canceled, Rejected
Vendor Service Type	Catering, Decor, Photography, Videography, Lighting, Stage Setup, Makeup Artist, DJ/Music, Transportation, Hosting/Anchor
Vendor Status	Available, Booked, Cancelled
Venue Availability Status	Available, Reserved, Not Available
Feedback Rating	1, 2, 3, 4, 5
Client Country	India, Philippines, Saudi Arabia, USA, Switzerland, Thailand, Maldives, Dubai, Sri Lanka, Bangkok, Singapore, Myanmar, France, Nepal

Event Budget formula field

Event Budget is a Formula field (return type Currency) that sets the budget from the event type:

```
CASE( TEXT(Event_Type__c),
  "Wedding", 50000, "Corporate", 30000, "Birthday", 10000,
  "Anniversary", 20000, "Festival", 60000, "Concert", 40000,
  "Other", 15000, 0 )
```

Relationships

Relationship	Type	Purpose
Event to Client	Lookup	Each event belongs to a client
Event to Venue	Lookup	Each event is held at a venue
Feedback to Event	Lookup	Feedback is given for an event
Feedback to Client	Lookup	Feedback comes from a client
Event Vendor to Event	Master-Detail	Junction side one
Event Vendor to Vendor	Master-Detail	Junction side two; gives the many-to-many link

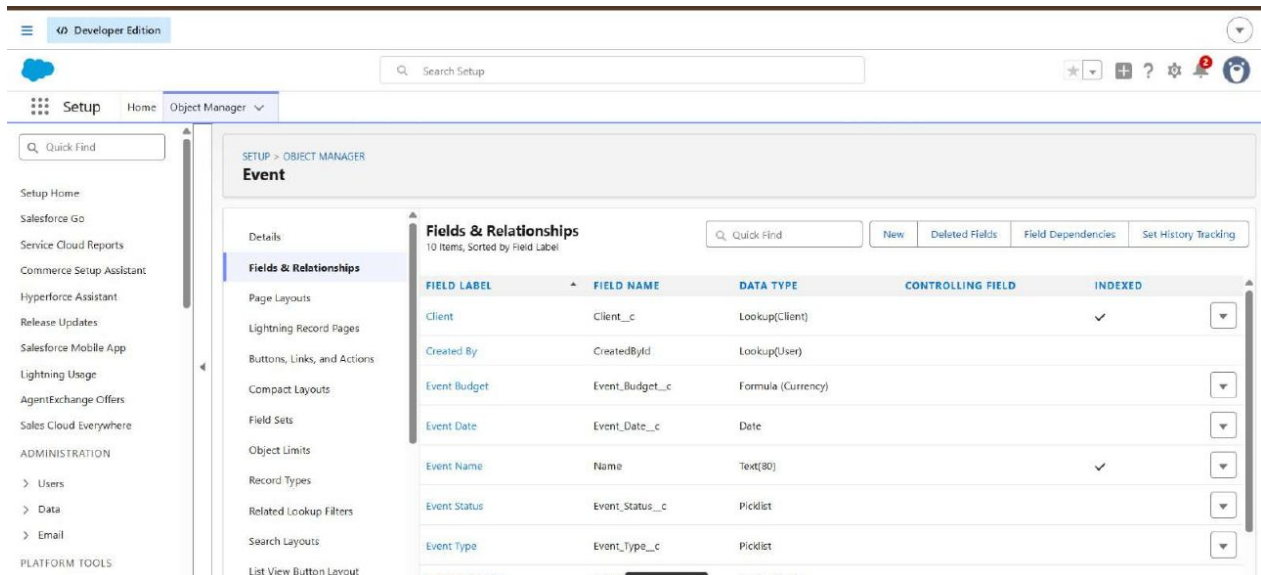


Figure 2 shows the Salesforce Setup interface for the Event object. The left sidebar lists various setup options, and the main content area displays the 'Fields & Relationships' section for the Event object. The table lists 10 fields, sorted by Field Label.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Client	Client__c	Lookup(Client)		✓
Created By	CreatedById	Lookup(User)		
Event Budget	Event_Budget__c	Formula (Currency)		
Event Date	Event_Date__c	Date		
Event Name	Name	Text(80)		✓
Event Status	Event_Status__c	Picklist		
Event Type	Event_Type__c	Picklist		

Figure 2. Event object: fields and relationships (Client lookup, Event Budget formula, picklists).

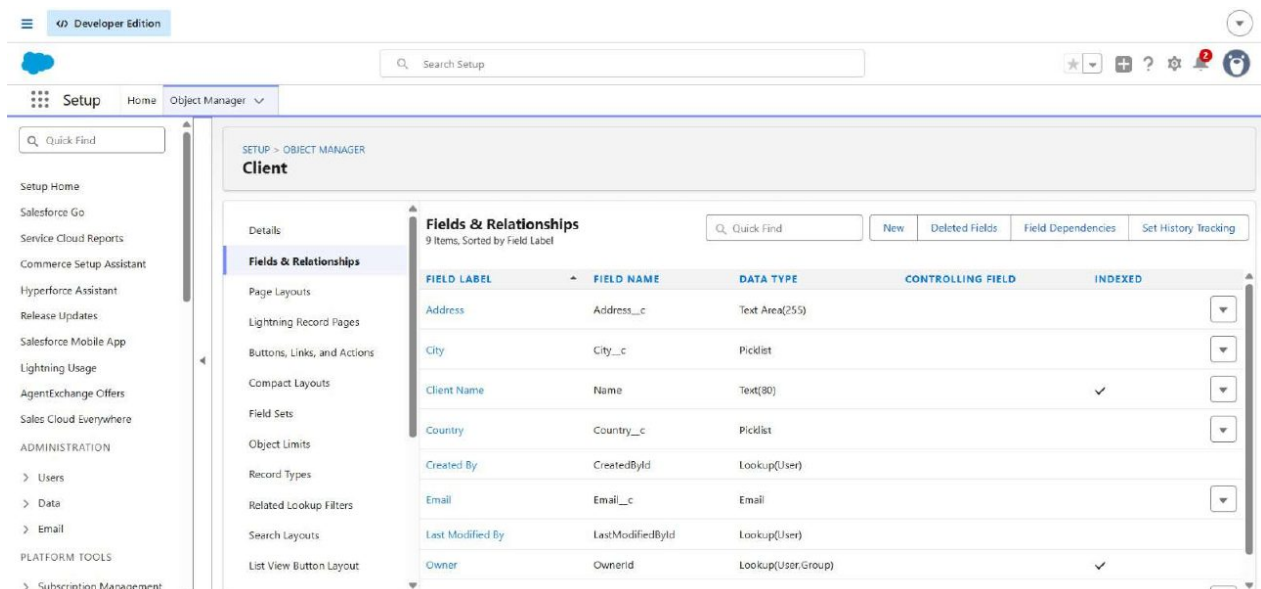


Figure 3 shows the Salesforce Setup interface for the Client object. The left sidebar lists various setup options, and the main content area displays the 'Fields & Relationships' section for the Client object. The table lists 9 fields, sorted by Field Label.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text Area(255)		
City	City__c	Picklist		
Client Name	Name	Text(80)		✓
Country	Country__c	Picklist		
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

Figure 3. Client object fields.

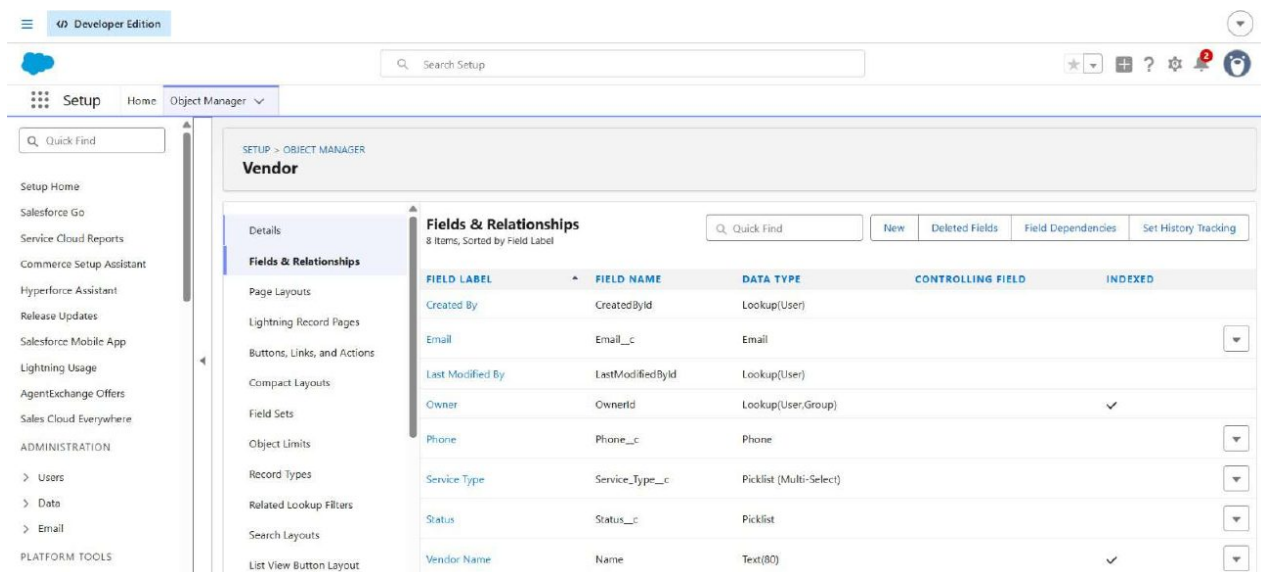


Figure 4 shows the Salesforce Setup interface for the Vendor object. The left sidebar lists various setup options, and the main content area displays the 'Fields & Relationships' section for the Vendor object. The table lists 8 fields, sorted by Field Label.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
Service Type	Service_Type__c	Picklist (Multi-Select)		
Status	Status__c	Picklist		
Vendor Name	Name	Text(80)		✓

Figure 4. Vendor object fields.

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar contains navigation links for Setup Home, Salesforce Go, Service Cloud Reports, Commerce Setup Assistant, Hyperforce Assistant, Release Updates, Salesforce Mobile App, Lightning Usage, AgentExchange Offers, Sales Cloud Everywhere, ADMINISTRATION (Users, Data, Email), PLATFORM TOOLS, and Subscription Management. The main content area is titled 'Venue' under 'Object Manager'. It displays a table of fields and relationships for the Venue object.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text Area(255)		
Availability Status	Availability__c	Picklist		
Capacity	Capacity__c	Number(18, 0)		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Location	Location__c	URL(255)		
Owner	OwnerId	Lookup(User,Group)		✓
Venue Name	Name	Text(30)		✓

Figure 5. Venue object fields.

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar contains navigation links for Setup Home, Salesforce Go, Service Cloud Reports, Commerce Setup Assistant, Hyperforce Assistant, Release Updates, Salesforce Mobile App, Lightning Usage, AgentExchange Offers, Sales Cloud Everywhere, ADMINISTRATION (Users, Data, Email), PLATFORM TOOLS, and Subscription Management. The main content area is titled 'Feedback' under 'Object Manager'. It displays a table of fields and relationships for the Feedback object.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Client	Client__c	Lookup(Client)		✓
Comments	Comments__c	Long Text Area(32768)		
Created By	CreatedById	Lookup(User)		
Event	Event__c	Lookup(Event)		✓
Feedback	Name	Auto Number		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Rating	Rating__c	Picklist		

Figure 6. Feedback object: Client and Event lookups with Rating and Comments.

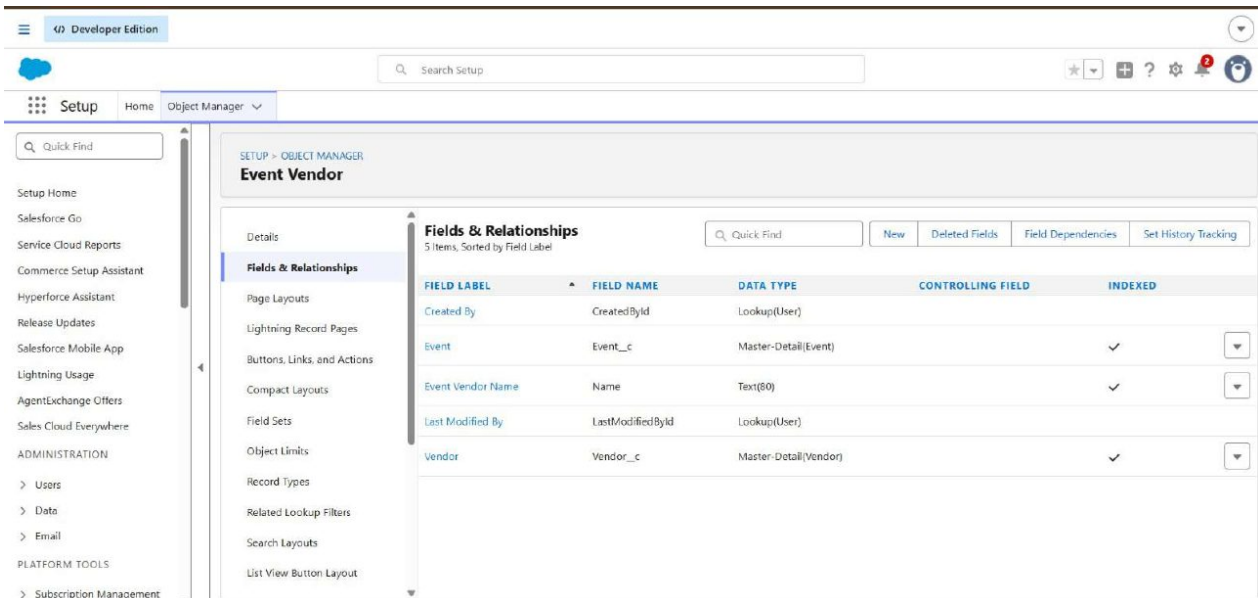


Figure 7. Event Vendor junction object with two Master-Detail relationships.

Lookup filter on Feedback

When a user creates Feedback and picks a Client, the Event lookup should show only that client's events. On the Feedback object, the Event lookup field was edited and a lookup filter was enabled: **Event: Client equals Feedback: Client.**

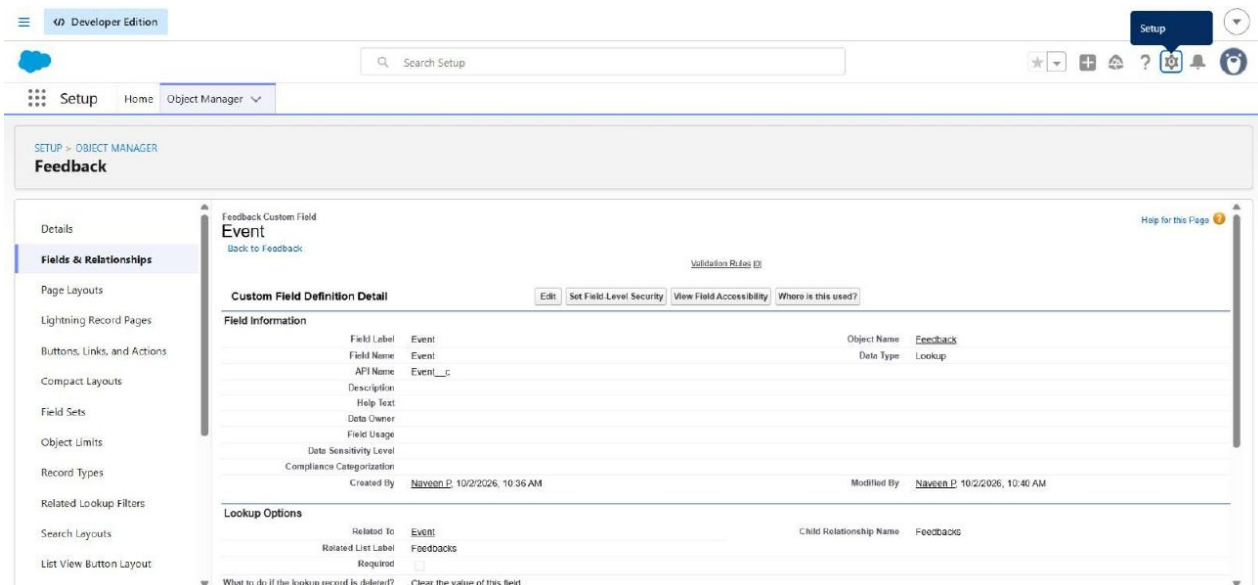


Figure 8. Feedback object: Event lookup field definition (lookup filter configured through Edit).

Milestone 4: Validation Rule

To keep contact data clean, the Client object has an active validation rule named **Email_Valid_Address**. It checks the Email field against a pattern and blocks the save when the format is wrong. The error message *Please Enter Valid Email Address* is shown on the Email field.

Error condition formula

NOT(REGEX(Email__c, "^[a-zA-Z0-9._]+@[a-zA-Z0-9._\.\.]{2,}\$"))

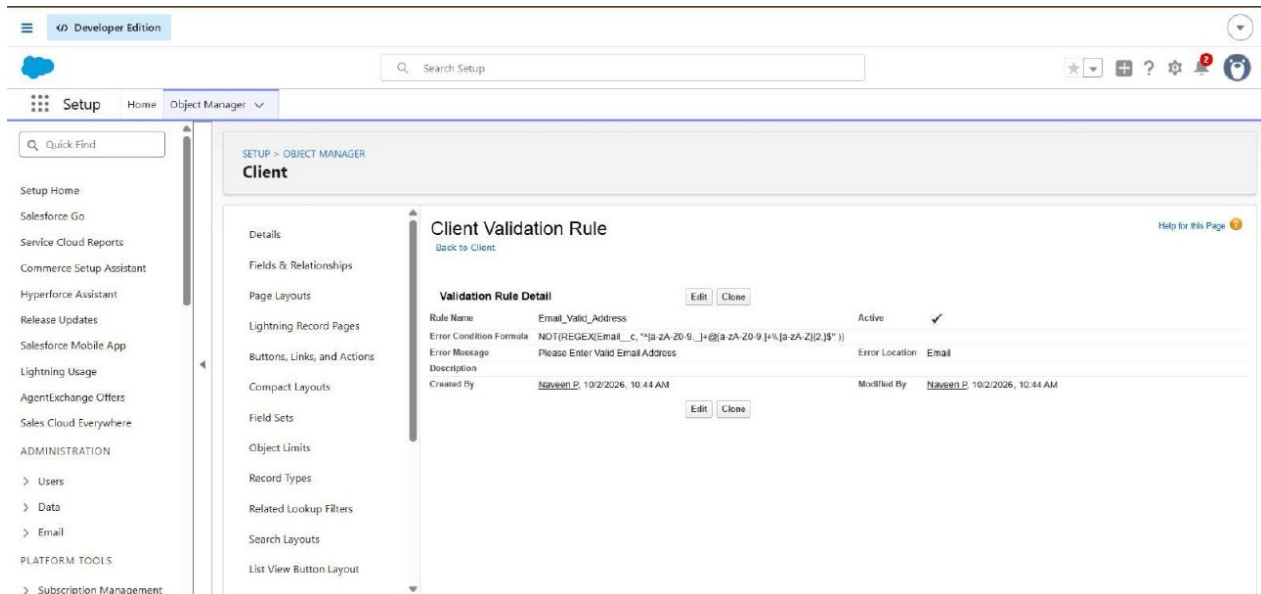


Figure 9. Client validation rule Email_Valid_Address.

Milestone 5: Approval Process for Event Cancellation

Before the approval process, four classic email templates were created (Setup > Email Templates):

Template	Subject	Used for
Event Cancellation Request Notification	Approval Request: Cancel Event {!Event__c.Name}	Approval assignment email
Email to Client After Cancellation	Event Cancellation Notice - {!Event__c.Name}	Informs the client
Approval Notification to Event Owner	Event Cancellation Approved: {!Event__c.Name}	Sent on approval
Rejection Notification to Event Owner	Event Cancellation Rejected: {!Event__c.Name}	Sent on rejection

The process **Event_Cancellation_Process** runs on the Event object. A record enters the approval when **Event Status equals Pending Cancellation**. The event owner is the initial submitter, the next approver is the manager of the record submitter, and a single step, **Manager Review**, sends the request for approval. The process is activated after configuration.

Stage	Actions configured
Initial submission	Record Lock; Email Alert "Alert - Event Cancellation Request"
Final approval	Field Update "Event Status Update" (status set to Canceled); Email Alert "Alert - Cancellation Approved" to the event owner
Final rejection	Field Update "Event Status To Rejected"; Email Alert "Alert - Cancellation Rejected" to the event owner; record unlocked
Recall	Record Unlock

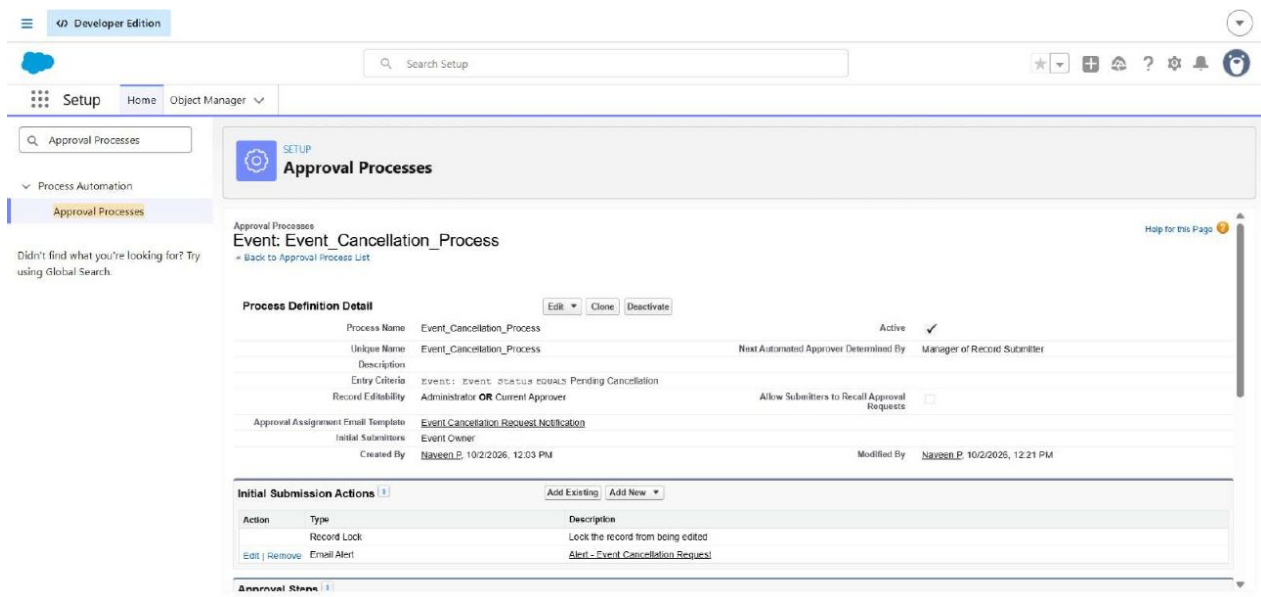


Figure 10. Approval Process detail: entry criteria, submitters and initial submission actions (process Active).

Milestone 6: Record-Triggered Flow

The Flow **Client Reminder - 3 Days Before** is a Record-Triggered Flow on the Event object. It starts when an Event is created or updated with **Event Status = Confirmed**. It has a **Run Immediately** path and a scheduled path, **3_Days_Before**, with the time source Event Date and an offset of 3 days before. The scheduled path runs the Send Email action **Alert_Client_3Day_Reminder**.

- Two text templates hold the message: a subject (*Reminder: Your Event {!\$Record.Name} is in 3 days*) and a body with client name, event name, date, venue and event type.
- The email goes to the Client's email address, with the event owner in CC.
- The Flow is saved and Active.

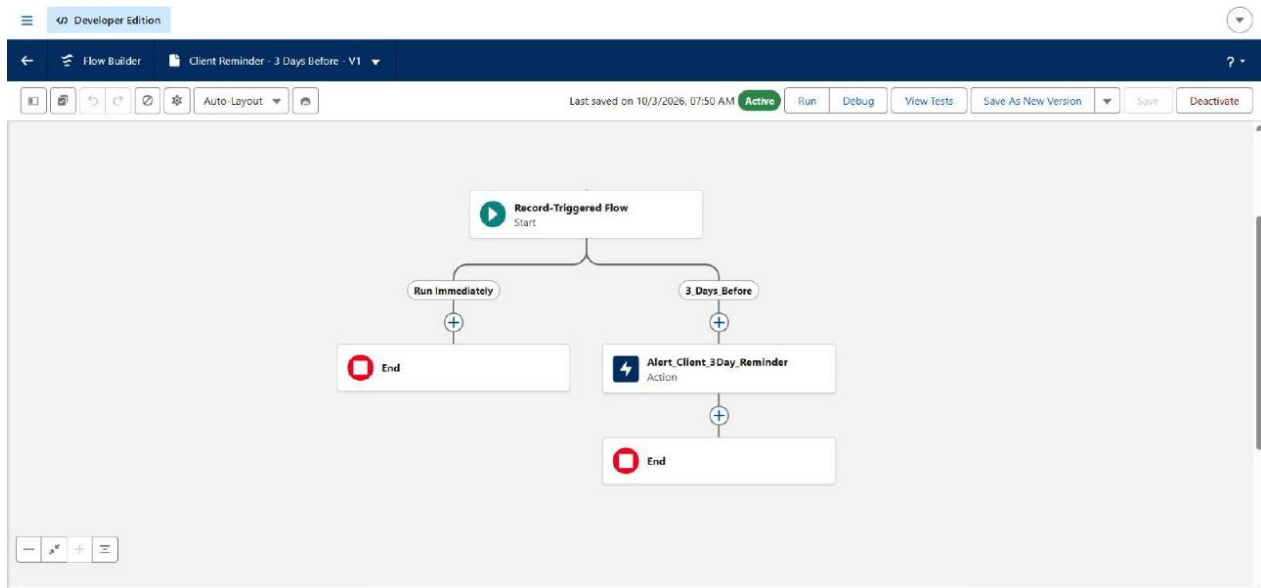


Figure 11. Flow Builder: Client Reminder - 3 Days Before (Active).

Milestone 7: Apex Classes and Triggers

Apex handles the business rules that need code. All code is written in API version 67 using the Developer Console.

Component	Type	What it does
EventTrigger13	Trigger on Event (after insert, after update)	Calls VenueStatusHelper whenever an event is saved.
VenueStatusHelper	Apex class (with sharing)	Sets the venue to Reserved when the event is Confirmed and back to Available when it is Canceled.
PreventDoubleBooking	Trigger on Event (before insert, before update)	Adds the error “This Venue is already booked on this date.” when another event uses the same venue and date.
BatchCompleteEvents	Batch Apex (Database.Batchable)	Finds events dated before today that are not Completed and marks them Completed.
ScheduleCompleteEvents	Schedulable Apex	Runs BatchCompleteEvents with a batch size of 200.

VenueStatusHelper.cls

```
public with sharing class VenueStatusHelper {
    public static void updateVenueStatus(List<Event__c> events){
        List<Venue__c> venues = new List<Venue__c>();
        for(Event__c ev : events){
            if(ev.Venue__c != null){
                if(ev.Event_Status__c == 'Confirmed'){
                    venues.add(new Venue__c(Id=ev.Venue__c, Availability__c='Reserved'));
                } else if(ev.Event_Status__c == 'Canceled'){
                    venues.add(new Venue__c(Id=ev.Venue__c, Availability__c='Available'));
                }
            }
        }
        if(!venues.isEmpty()) update venues;
    }
}
```

EventTrigger13.trigger

```
trigger EventTrigger13 on Event__c (after insert, after update) {
    VenueStatusHelper.updateVenueStatus(Trigger.new);
}
```

PreventDoubleBooking.trigger

```
trigger PreventDoubleBooking on Event__c (before insert, before update) {
    Set<Id> venueIds = new Set<Id>();
    for(Event__c ev : Trigger.new){
        if(ev.Venue__c != null) venueIds.add(ev.Venue__c);
    }
    Map<Id, List<Event__c>> venueEventMap = new Map<Id, List<Event__c>>();
    for(Event__c e : [SELECT Id, Venue__c, Event_Date__c FROM Event__c WHERE Venue__c IN :venueIds]){
        if(!venueEventMap.containsKey(e.Venue__c)){
            venueEventMap.put(e.Venue__c, new List<Event__c>());
        }
        venueEventMap.get(e.Venue__c).add(e);
    }
    for(Event__c ev : Trigger.new){
        if(ev.Venue__c != null && venueEventMap.containsKey(ev.Venue__c)){
            for(Event__c existing : venueEventMap.get(ev.Venue__c)){
                if(existing.Event_Date__c == ev.Event_Date__c && existing.Id != ev.Id){
                    ev.addError('This Venue is already booked on this date.');
```

BatchCompleteEvents.cls

```
global class BatchCompleteEvents implements Database.Batchable<sObject> {
    global Database.QueryLocator start(Database.BatchableContext bc){
        return Database.getQueryLocator(
            'SELECT Id, Event_Date__c, Event_Status__c FROM Event__c ' +
            'WHERE Event_Date__c < TODAY AND Event_Status__c != \'Completed\'' );
    }
    global void execute(Database.BatchableContext bc, List<Event__c> scope){
        for(Event__c ev : scope){ ev.Event_Status__c = 'Completed'; }
        update scope;
    }
    global void finish(Database.BatchableContext bc){
        System.debug('Past events marked as Completed');
    }
}
```

ScheduleCompleteEvents.cls

```
global class ScheduleCompleteEvents implements Schedulable {
    global void execute(SchedulableContext sc){
        Database.executeBatch(new BatchCompleteEvents(), 200);
    }
}
```

Scheduling the batch job

From Setup > Apex Classes > Schedule Apex: job name **Daily Event Completion**, class

ScheduleCompleteEvents, frequency Weekly with all days selected, preferred start time 8:00 PM. Salesforce then marks past events as Completed every evening.

Apex Classes

Apex Code is an object oriented programming language that allows developers to develop on-demand business applications on the Lightning Platform.

Percent of Apex Used: 0.02%
You are currently using 952 characters of Apex Code (excluding comments and @isTest annotated classes) in your organization, out of an allowed limit of 6,000,000 characters. Note that the amount in use includes both Apex Classes and Triggers defined in your organization.

Estimate your organization's code coverage [1]
Compute all classes [1]

View: [All v] Create New Idea

Action	Name	Namespace Prefix	API Version	Status	Size Without Comments	Last Modified By	Has Trace Flag
Edit Del Security	BatchCompleteEvents		67.0	Active	651	Navleen P 10/2/2026, 7:39 PM	☐
Security	DeveloperEditionUtils	devedapp	64.0	Active	164	OrgFarm EPIC, 9/29/2026, 9:23 AM	☐
Security	DeveloperEditionUtilsTest	devedapp	64.0	Active	261	OrgFarm EPIC, 9/29/2026, 9:23 AM	☐
Security	PostinstallScript	devedapp	64.0	Active	2,175	OrgFarm EPIC, 9/29/2026, 9:23 AM	☐
Security	PostinstallScriptTest	devedapp	64.0	Active	781	OrgFarm EPIC, 9/29/2026, 9:23 AM	☐
Edit Del Security	ScheduleCompleteEvents		67.0	Active	160	Navleen P 10/2/2026, 7:40 PM	☐
Edit Del Security	VenueStatusHelper		67.0	Active	35	Navleen P 10/2/2026, 7:26 PM	☐

Figure 12. Apex Classes list in Setup: BatchCompleteEvents, ScheduleCompleteEvents and VenueStatusHelper (API 67.0, Active).

Milestone 8: Custom Tabs

A custom tab was created for each object (Setup > Tabs > Custom Object Tabs > New) so that users can reach records quickly: **Clients, Events, Feedbacks, Vendors and Venues**.

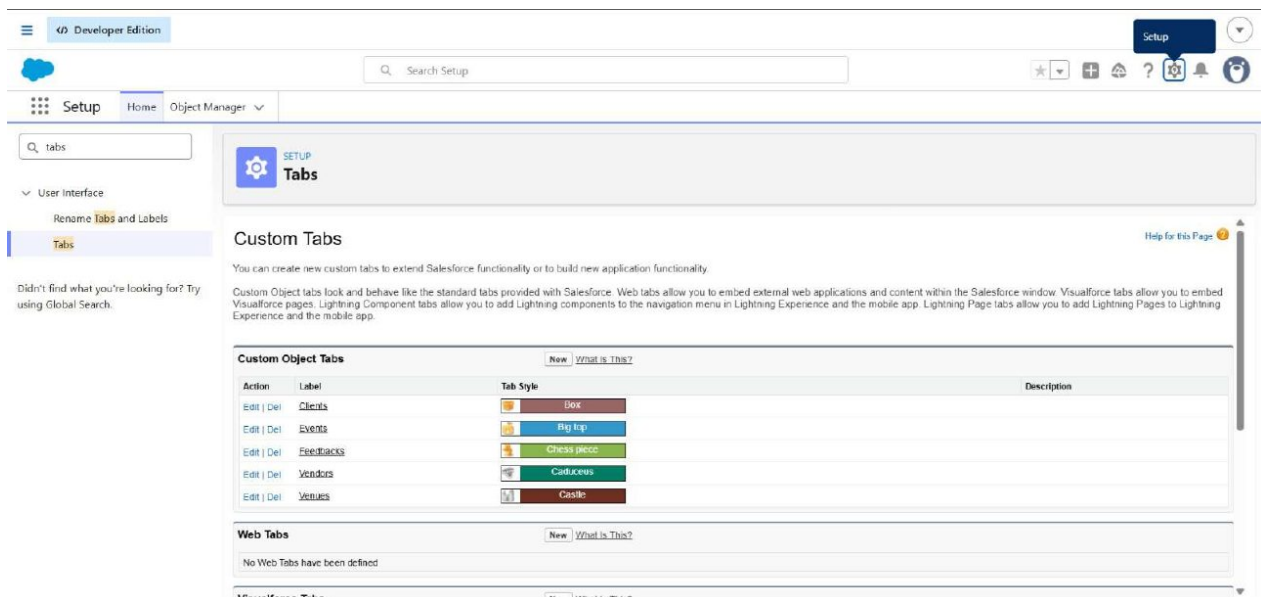


Figure 13. Custom Object Tabs in Setup.

Milestone 9: Lightning App

The Lightning App **Event Planner** groups the project in one place (Setup > App Manager > New Lightning App). Its navigation bar contains Events, Clients, Vendors, Venues, Feedbacks, Reports and Dashboards, so a user does not need to search Setup to do daily work. The app was made available to the required profiles.

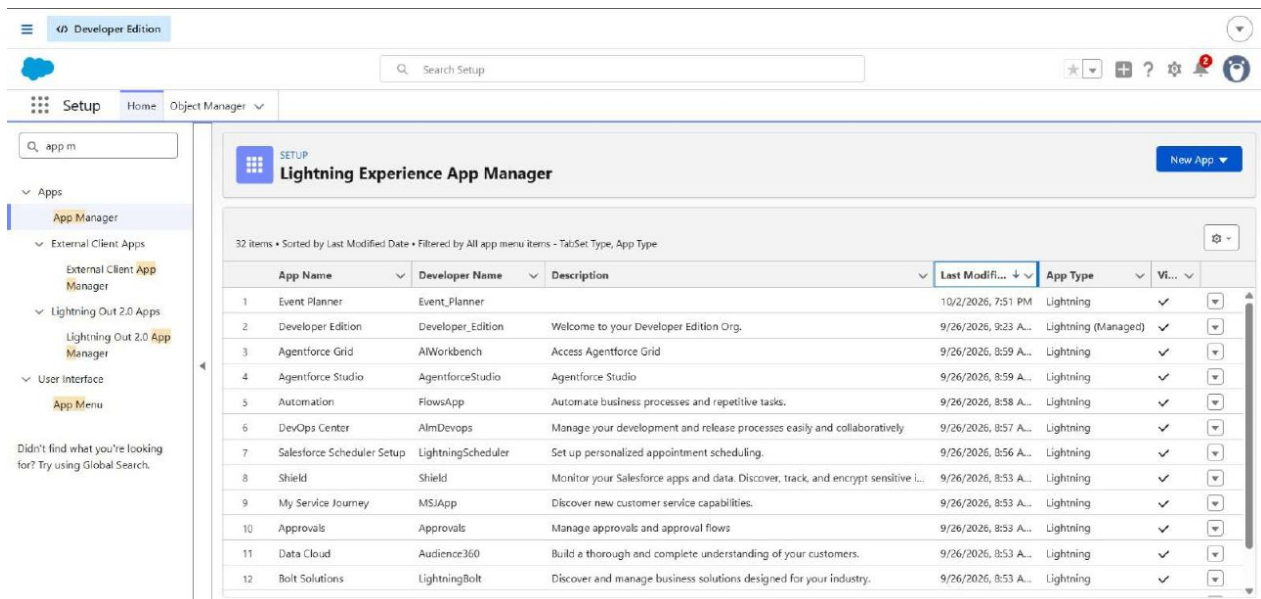


Figure 14. Lightning Experience App Manager showing the Event Planner app.

Milestone 10: Reports

A report folder named **EventForce Report** was created and shared by role (Event Coordinator and Vendor Manager with View access). Records were added to every object first so that the reports have data to show.

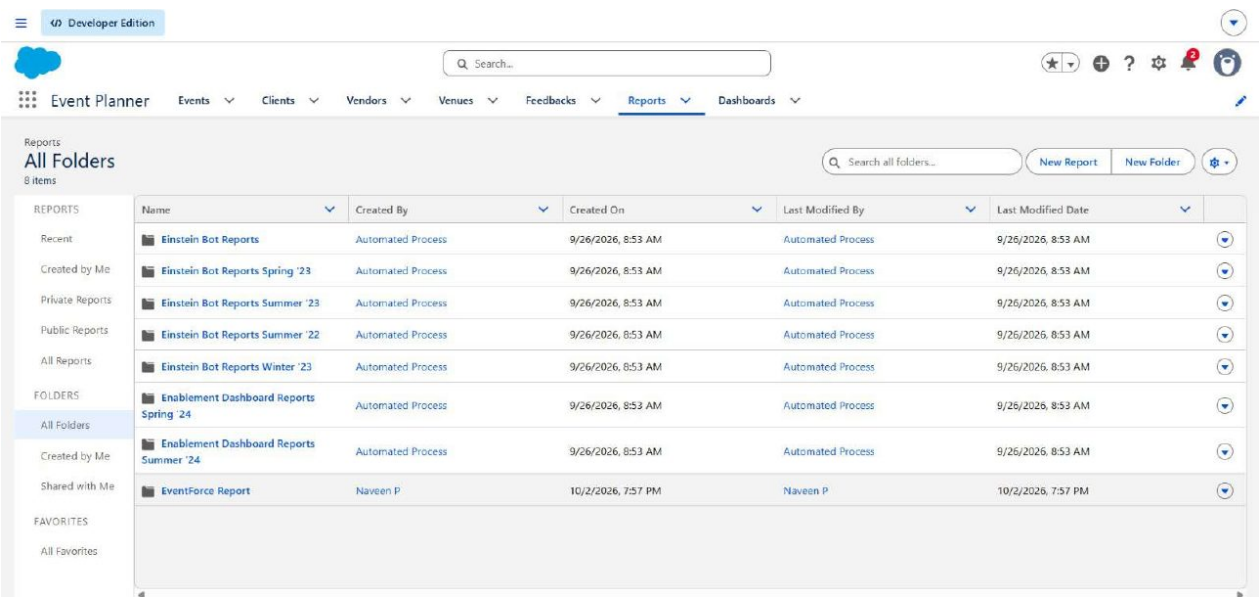


Figure 15. Reports tab in the Event Planner app with the EventForce Report folder.

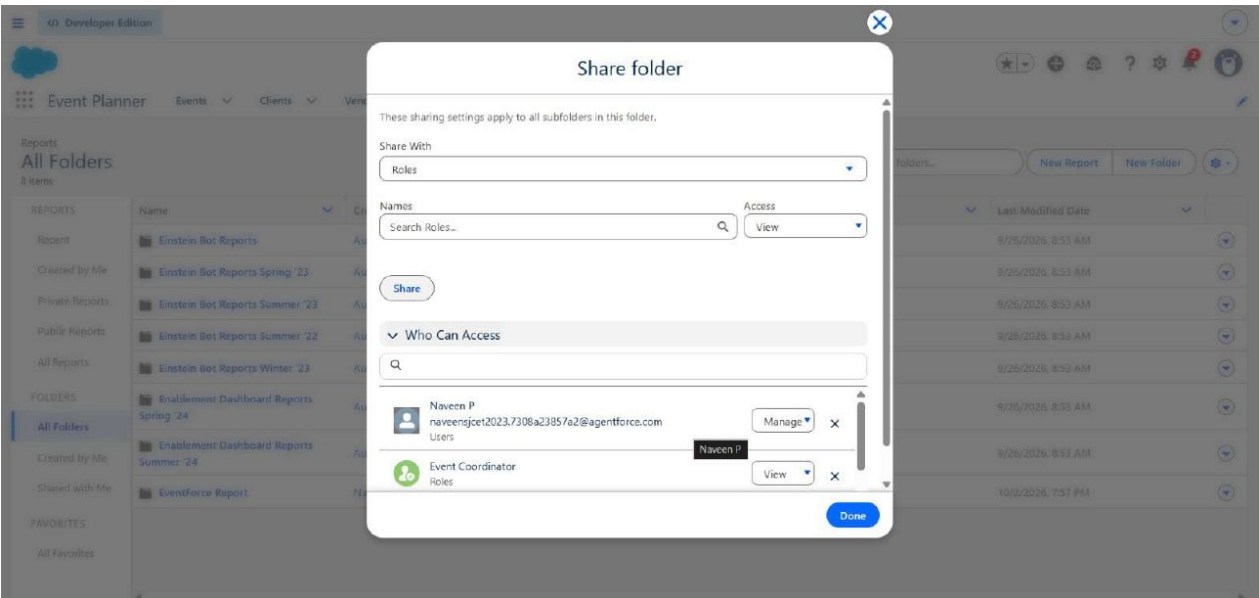


Figure 16. Sharing the report folder with roles (Event Coordinator: View).

Report 1: New Events with Venue Report. Uses the Events with Venue report type and shows Event Name, Event Date, Venue Name, Event Type, Client and Event Status for each event.

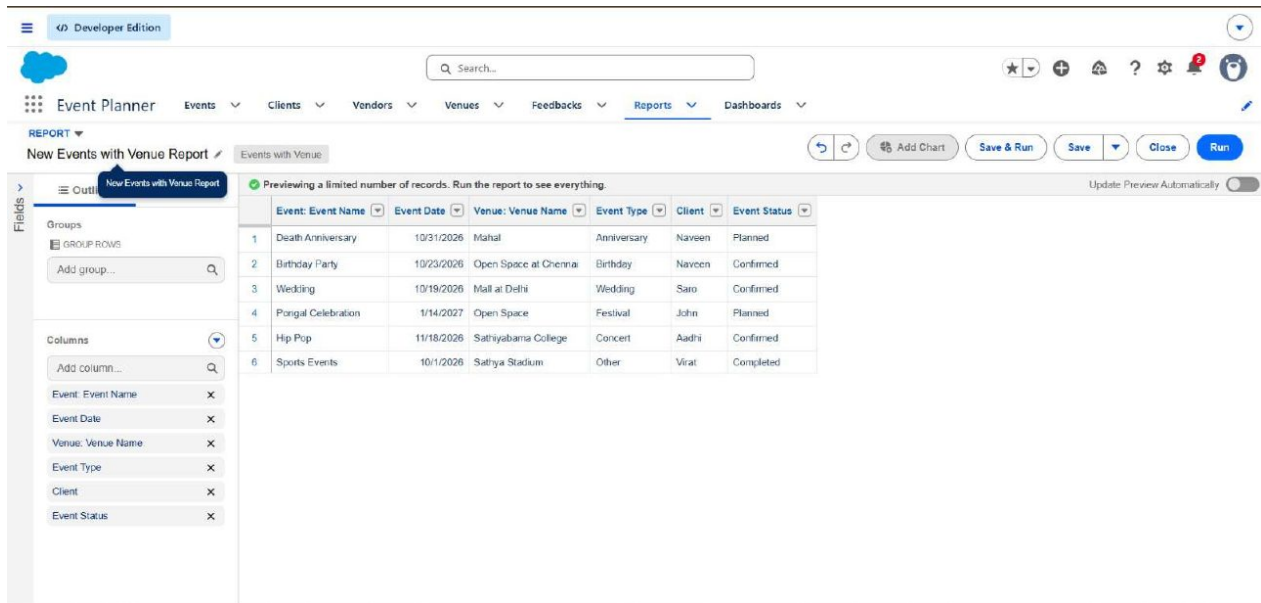


Figure 17. New Events with Venue Report (report builder).

Report 2: Upcoming Events by Month. Rows are grouped by Event Date (calendar month) with a record count and a chart, and the columns include Event Name, Venue Name, Event Budget, Event Type, Client and Event Status.

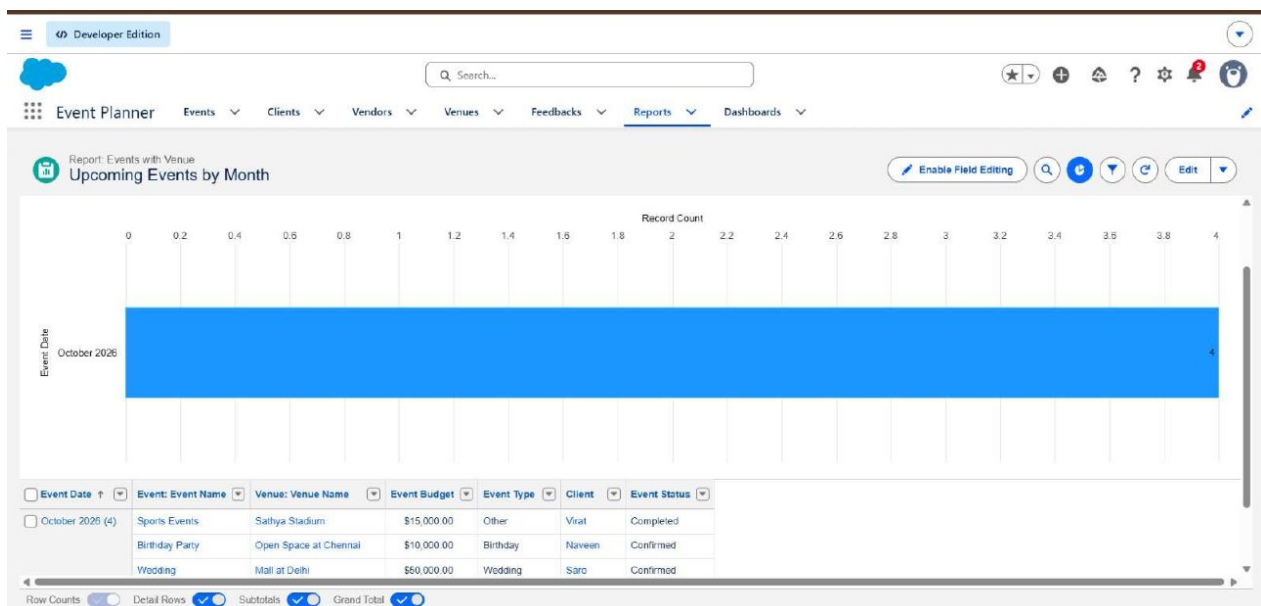


Figure 18. Upcoming Events by Month report with chart.

Milestone 11: Dashboard

The dashboard **EventForce Operations Dashboard** shows the donut chart *Upcoming Events by Month* (subtitle: Upcoming Events Overview), built from the report data. It gives coordinators a quick view of how many events are coming up.

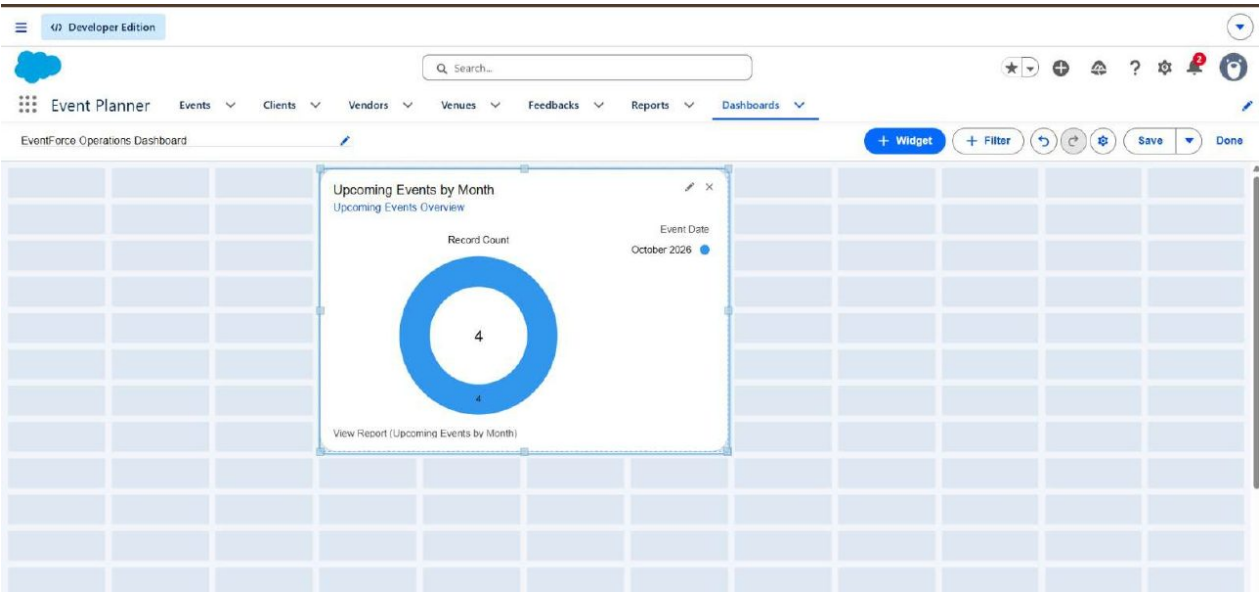


Figure 19. EventForce Operations Dashboard with the Upcoming Events by Month donut chart.

Milestone 12: Data Migration

Records for Events, Clients, Vendors, Venues and Feedback were loaded with the **Data Import Wizard** after pre-validation checks to avoid duplicates and keep the data consistent.

Milestone 13: Profiles

Custom profiles control what each user type can see and do. The profiles are **Event Admin** (cloned from System Administrator, so it inherits full access), **Event Coordinator**, **Vendor Manager** and **Client Profile** (the last three cloned from Standard Platform User with the Salesforce Platform license). Event Planner is the default app for them, the session timeout is 2 hours, passwords never expire and the minimum password length is 8.

Profile	Event	Client	Vendor	Venue	Feedback
Event Admin	Full	Full	Full	Full	Full
Event Coordinator	Create, Read, Edit, Delete	Read, Edit	Read	Read	Create, Read
Vendor Manager	Read	Read	Create, Read, Edit, Delete	Read	Read
Client Profile	Read (own events)	-	No access	No access	Create, Read

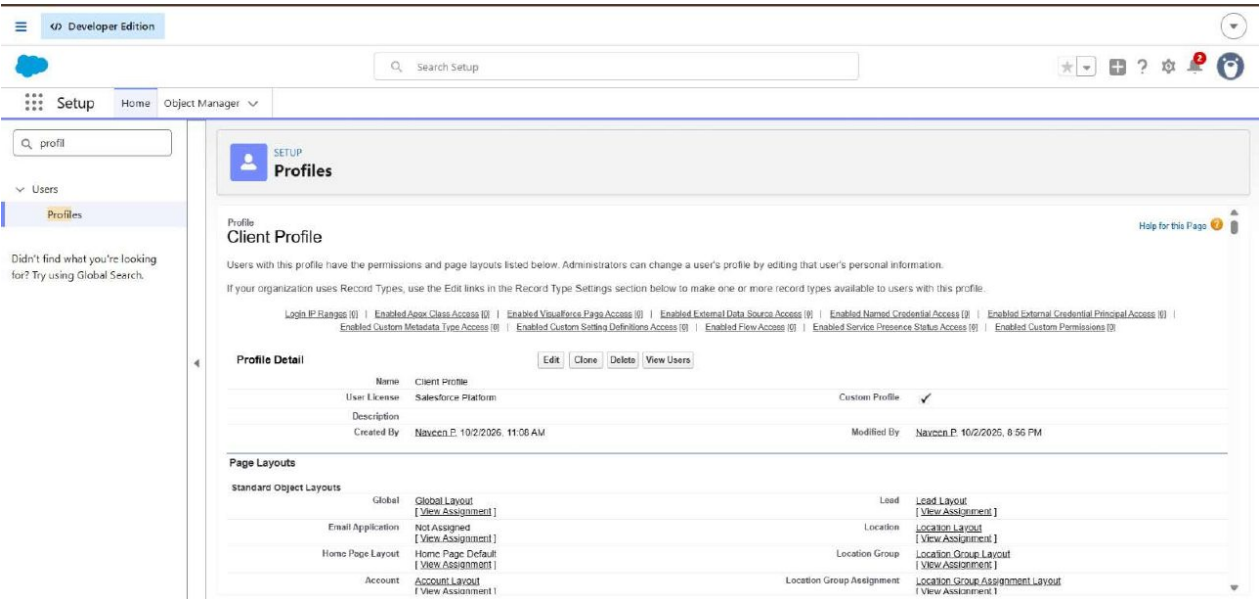


Figure 20. Client Profile detail (custom profile, Salesforce Platform license).

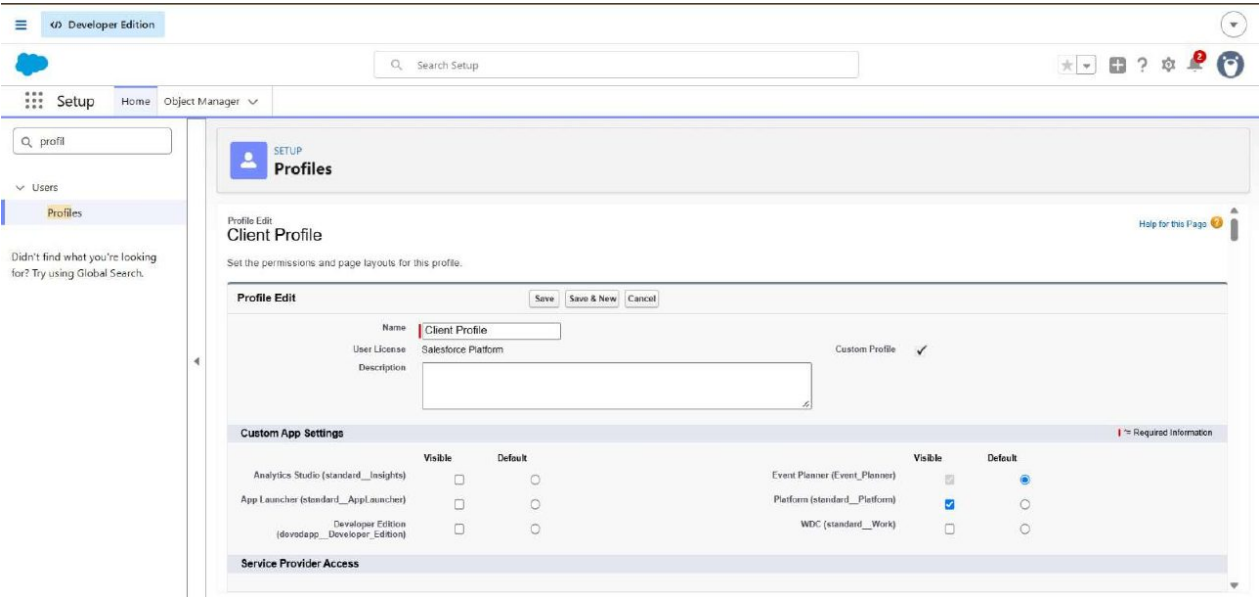


Figure 21. Client Profile edit: Event Planner set as the default custom app.

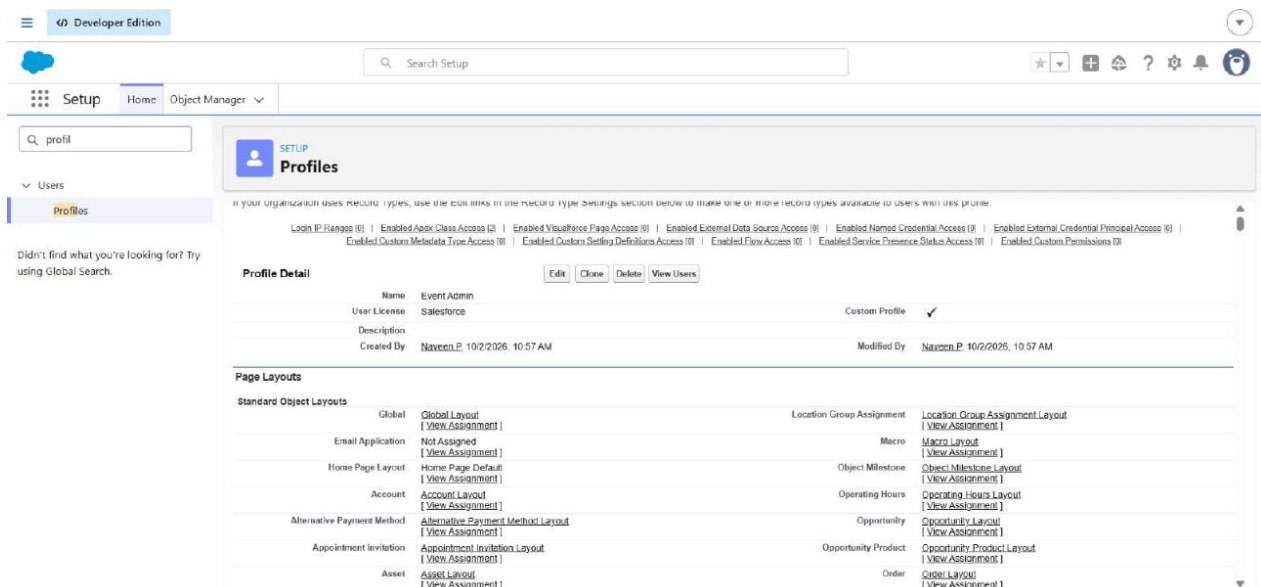


Figure 22. Event Admin profile (custom profile, Salesforce license).

Milestone 14: Roles

The role hierarchy places **Event Admin** under the CEO. The roles **Client**, **Event Coordinator** and **Vendor Manager** sit below Event Admin, so records owned by them are visible upward in the hierarchy.

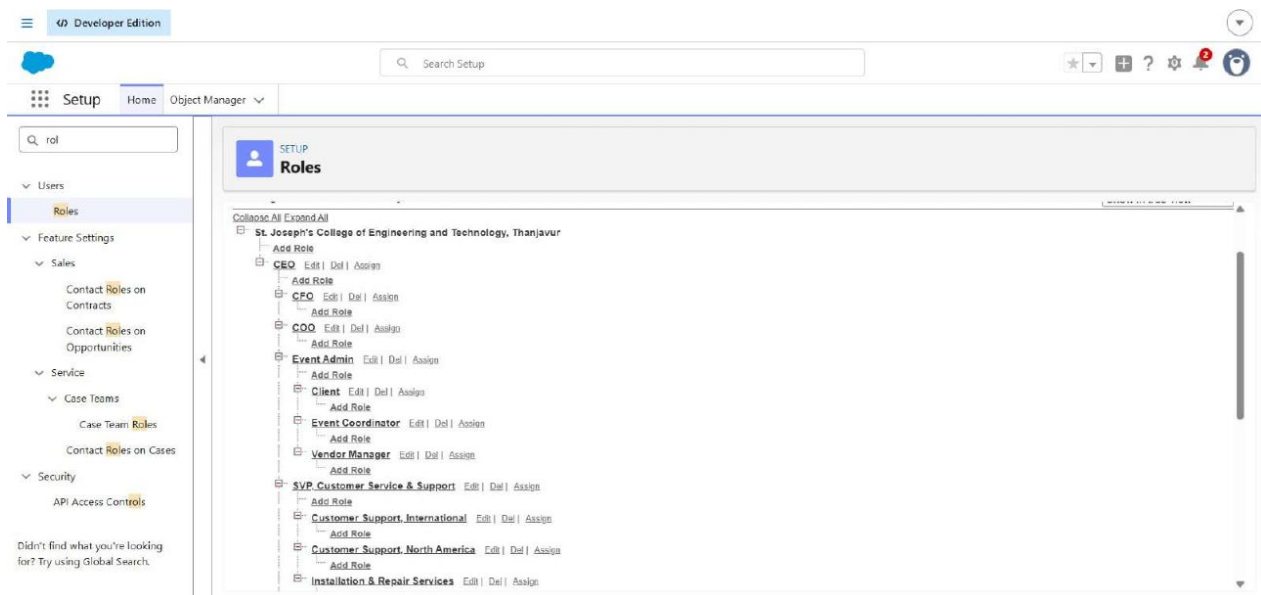
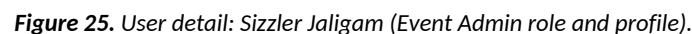
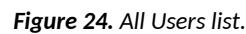


Figure 23. Role hierarchy with Event Admin and the three roles below it.

Milestone 15: Users

User	Role	Profile	License
Sizzler Jaligam	Event Admin	Event Admin	Salesforce
Roopan G	Event Coordinator	Event Coordinator	Salesforce Platform
Saravanan S	Vendor Manager	Vendor Manager	Salesforce Platform



The permission set **Feedback Manager** adds extra access on top of a profile without changing the profile itself. In Object Settings, the Feedback object was given Read, Create, Edit and Delete. The set is assigned to the user **Roopan G** (Event Coordinator).

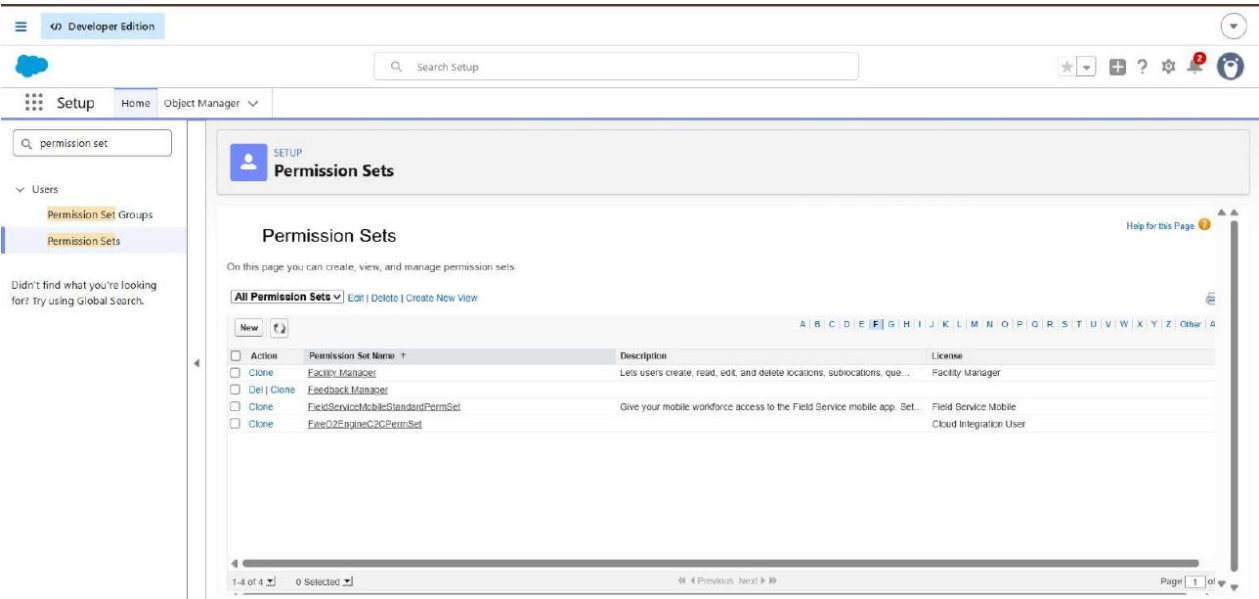


Figure 26. Permission Sets list showing Feedback Manager.

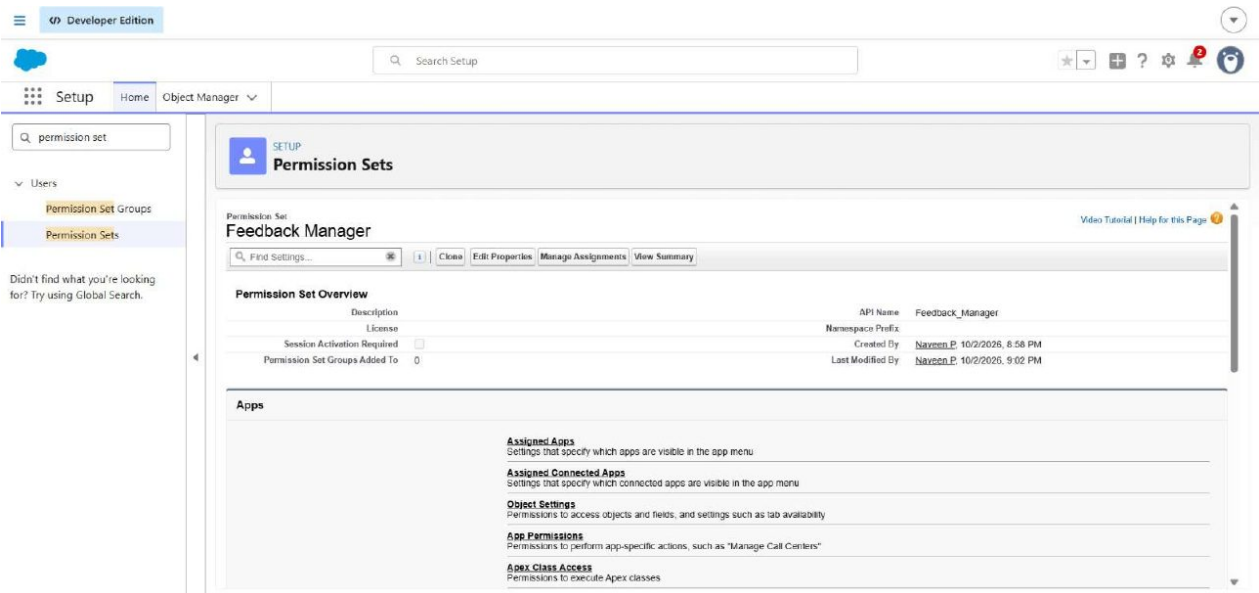


Figure 27. Feedback Manager permission set overview.

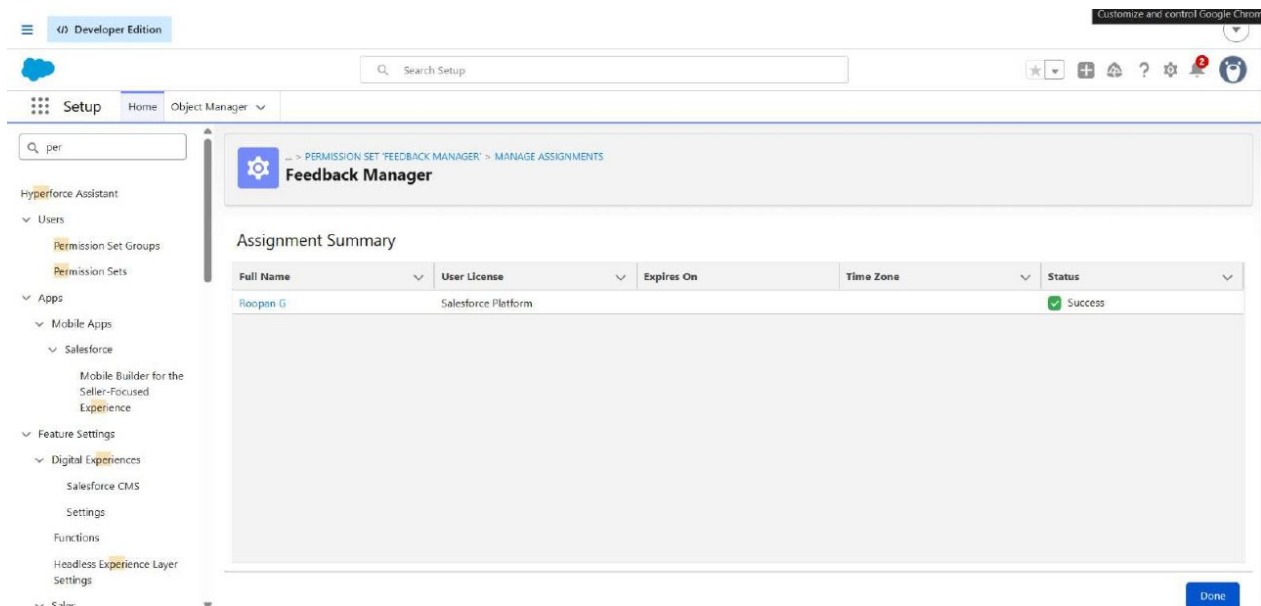


Figure 28. Assignment summary: Feedback Manager assigned to Roopan G.

Milestone 17: Sharing Settings (OWD)

Organization-Wide Defaults set the baseline record access. **Event is Private**, so users see only their own events and those of users below them in the role hierarchy. **Client, Feedback, Vendor and Venue are Public Read/Write** because these are shared reference records. “Grant Access Using Hierarchies” is enabled for the custom objects.

Object	Default internal access	Default external access
Event	Private	Private
Client	Public Read/Write	Private
Feedback	Public Read/Write	Private
Vendor	Public Read/Write	Private
Venue	Public Read/Write	Private

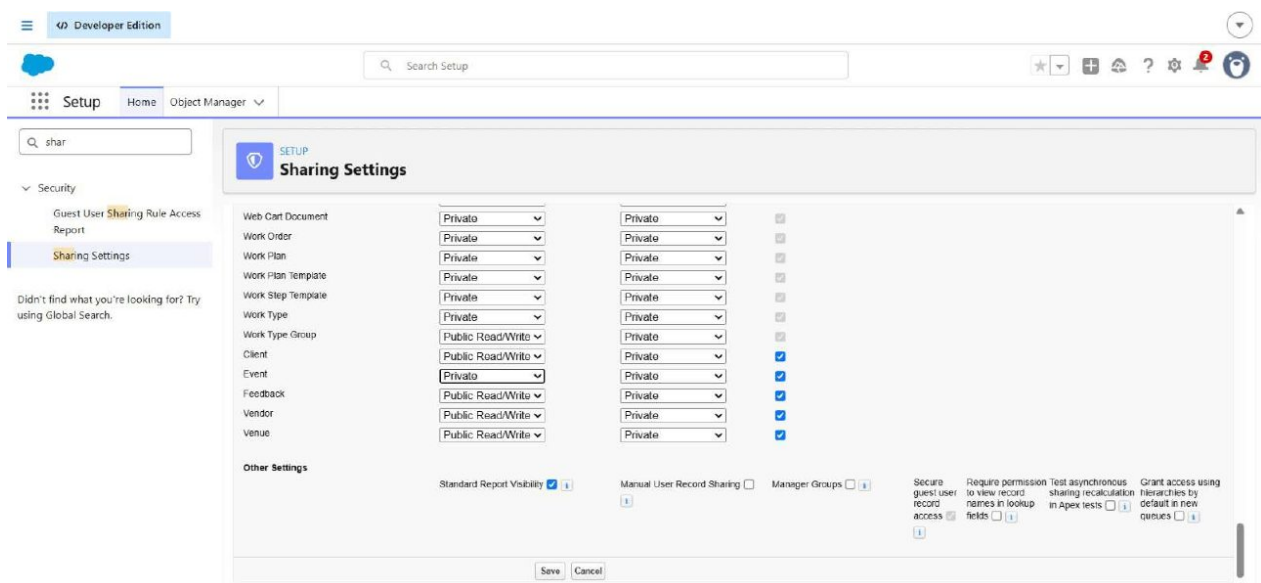


Figure 29. Organization-Wide Defaults for the custom objects (Event: Private).

Milestone 18: Sharing Rule

Because Event is Private, a criteria/owner-based sharing rule **Event_Sharing_For_Vendors** shares events owned by the **Event Coordinator** role with the **Vendor Manager** role at **Read Only** access, so vendor managers can see the events they need to supply.

7. Functional and Performance Testing

The table lists the checks planned for the main features. The Result column is filled in after each test is run in the org, with a screenshot kept as evidence.

ID	Feature	Test steps	Expected result	Result
TC-01	Email validation	Create a Client with an invalid email such as "abc@".	Save is blocked with "Please Enter Valid Email Address".	Pending
TC-02	Double booking	Create two Events at the same Venue on the same date.	Second save fails with "This Venue is already booked on this date."	Pending
TC-03	Venue reserved	Set an Event to Confirmed.	Venue Availability Status becomes Reserved.	Pending
TC-04	Venue available	Set the Event to Canceled.	Venue Availability Status becomes Available.	Pending
TC-05	Cancellation approval	Set Event Status to Pending Cancellation, submit for approval and approve it.	Record locks on submit; on approval the status becomes Canceled and an email alert is sent.	Pending
TC-06	Cancellation rejection	Repeat TC-05 and reject the request.	Record unlocks, status changes to Rejected, rejection email is sent.	Pending
TC-07	Client reminder Flow	Create a Confirmed Event dated a few days ahead.	Scheduled path sends the reminder 3 days before the Event Date.	Pending
TC-08	Batch job	Run BatchCompleteEvents with a past, non-completed Event.	The past Event is set to Completed.	Pending
TC-09	Access control	Log in as Event Coordinator and Vendor Manager.	Each user sees only what the profile, role and OWD allow.	Pending
TC-10	Sharing rule	As Vendor Manager, open an Event owned by an Event Coordinator.	The Event is visible with Read Only access.	Pending

8. Advantages and Disadvantages

Advantages

- All event, client, venue, vendor and feedback data sits in one connected CRM.
- A venue cannot be double booked on the same date; the error is shown before the save.
- Venue availability is updated by code, with no manual step.
- Clients are reminded three days before their event by the Flow, which reduces manual effort.
- Cancellations follow a manager approval with record lock and email alerts.
- Reports and a dashboard give visibility into upcoming events.
- Profiles, roles, a permission set, OWD and a sharing rule keep access under control.
- Validation rules and pre-checked data loading keep records accurate.
- A batch job marks past events as Completed.

Disadvantages

- The double-booking check compares only the event date, not start and end times.
- Venue status follows the last saved event, so cancelling one event can show a venue as Available even if it has another confirmed event on a different date.
- The approval has a single step with one approver.
- No Apex test classes are written yet, which a production deployment would need.
- The project was built in a Developer Edition org for academic purposes and was not deployed to production.
- Real deployment needs sandboxes, Change Sets, Salesforce DX or CI/CD tools.
- Flows, Apex jobs, approvals and sharing need regular monitoring and maintenance.
- Reminder emails depend on a verified email sending domain.

9. Conclusion

EventForce shows how a standard Salesforce org can be shaped into a working event management system. Events, Clients, Vendors, Venues, Feedback and Event Vendor data now live on one platform. With a clear data model, the system blocks double bookings, keeps venue availability current, reminds clients before their event, routes cancellations for approval and limits access by role. Through this project the team learned object design and relationships, Flow with scheduled paths, approval processes, Batch and Schedulable Apex, data migration and the Salesforce security model. The outcome is less manual follow-up, cleaner data and a clear view of upcoming events for the whole team.

10. Future Scope

- **Production deployment:** move the solution from the Developer org to a production org using sandboxes and proper deployment tools.
- **Smarter booking:** check time slots and overlapping dates, not only the event date.
- **Approvals:** add more approval steps and escalation when a request is not answered.
- **Vendor management:** add vendor tracking and performance management.
- **Analytics:** add reports and dashboards for vendor performance, feedback ratings and budgets.
- **Integrations and mobile:** connect external services (for example SMS or WhatsApp reminders) and improve the mobile experience.
- **Quality:** write Apex test classes with enough code coverage for a production release.
- **Maintenance:** regularly review Flows, Apex, validation rules and sharing settings as requirements change.

11. Appendix

Ref	Item	Description
A	Custom objects	Event__c, Client__c, Vendor__c, Venue__c, Feedback__c, Event Vendor (junction)
B	Validation rule	Email_Valid_Address on Client
C	Approval Process	Event_Cancellation_Process on Event
D	Flow	Client Reminder - 3 Days Before (Record-Triggered, scheduled path)
E	Apex	VenueStatusHelper, EventTrigger13, PreventDoubleBooking, BatchCompleteEvents, ScheduleCompleteEvents
F	Lightning App	Event Planner
G	Reports and Dashboard	New Events with Venue Report, Upcoming Events by Month, EventForce Operations Dashboard
H	Security	Event Admin, Event Coordinator, Vendor Manager, Client Profile; Feedback Manager permission set; OWD; Event_Sharing_For_Vendors
I	Data migration	Salesforce Data Import Wizard

12. Demo Link

Demo video:

https://drive.google.com/file/d/1NpV4C8CTLPUhZ93tmmTgQ7dJTvLAX6D/view?usp=drive_link

Project documentation:

https://drive.google.com/file/d/1amsxcijxsh2KM8-w_aUUmD6-F6Mo8gEC/view?usp=drive_link